AgroAware: A Generative AI-Based Farming Advisor and Awareness Platform

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***Abstract*—Indian agriculture is increasingly shifting toward data-driven practices, yet small and marginal farmers still rely on intuition-based decision-making for crop selection and fertilizer planning. This results in inconsistent yields and low resilience to climatic variability. Recent advances in Machine Learning (ML) and Generative AI (GenAI) provide opportunities to merge predictive intelligence with accessible advisory communication.**

AgroAware introduces a unified ML–GenAI framework that integrates an ensemble-based crop prediction model with a multilingual advisory engine. The ensemble aims to utilize algorithms such as Decision Tree, Random Forest, Logistic Regression, and XGBoost to analyze soil nutrients (N, P, K), pH, rainfall, temperature, and humidity for accurate crop checks. A Retrieval-Augmented Generation (RAG) module combined with a fine-tuned LLaMA model is proposed to transform predictions into context-aware advisories in regional languages.

By addressing gaps in linguistic accessibility and farmer awareness, AgroAware bridges computational intelligence with practical usability. The proposed framework aims to demonstrate improved inclusivity and applicability across diverse agricultural settings.

***Index Terms*—Agriculture, Machine Learning, Ensemble Learning, Generative AI, RAG, LLaMA, Precision Farming, Explainable AI**

I. Introduction

Agriculture remains central to India's economy, supporting over half of the population. Yet most small and marginal farmers continue to rely on traditional, intuition-based practices for crop selection, fertilizer planning, and risk assessment. These non-scientific approaches often ignore soil nutrient profiles, rainfall variation, and climatic factors, leading to inconsistent yields and economic vulnerability [1], [2]. Although AI and Machine Learning (ML) have advanced significantly—achieving high accuracy in disease detection [1], [3], [15], crop recommendation [5], [10], [12], and yield prediction [3]—their real-world adoption remains limited due to challenges in accessibility and linguistic suitability. Furthermore, a significant "digital divide" persists, where small-scale farmers lack access to real-time data interpretation, leaving them dependent on static, generalized advice rather than dynamic, field-specific insights.

Parallel developments in Generative AI (GenAI) and large language models (LLMs) have created new possibilities for agricultural communication. Systems such as multilingual LLaMA-based advisories [7] and conversational chatbots [13], [14] demonstrate strong potential to simplify complex agricultural information. However, these solutions typically operate independent of predictive ML models, resulting in generalized advisories that lack scientific grounding.

A. Motivation

The motivation behind **AgroAware** arises from the disconnect between highly accurate ML models and farmer-friendly communication systems. While ML frameworks provide numerical predictions [5], [10], [12], farmers often lack the scientific context to trust or apply these results. Conversely, GenAI-based advisory systems [7], [13] offer linguistic accessibility but lack real-time, data-driven insights. Further, digital agriculture platforms face adoption barriers owing to limited literacy, language diversity, and infrastructural gaps in rural regions [8], [9], [11].

To address these challenges, AgroAware proposes to integrate a robust ML ensemble with multilingual generative intelligence, enabling farmers to receive interpretable and context-aware recommendations.

B. Problem Definition

Agriculture faces three interlinked gaps:

1) **Predictive Gap:** ML models often lack explainability and fail to provide interpretable crop recommendations [5], [10], [12].
2) **Awareness Gap:** Farmers are not equipped with accessible guidelines on fertilizer use, soil constraints, or climatic risks [7], [13].
3) **Communication Gap:** Most systems deliver advisories in English or technical terminology, limiting reach in multilingual rural settings [7], [14].

These gaps reduce trust in AI-based systems and hinder adoption in real-world farming.

C. Research Objectives

AgroAware is developed to unify prediction, explanation, and awareness generation through:

- To develop a machine learning based crop recommendation system that analyzes soil nutrients, pH, and weather conditions to suggest the most suitable crops.
- To build a multilingual user interface (Kannada, Hindi, Telugu, English) to ensure accessibility for farmers from diverse language backgrounds.
- To create both text-based and voice-enabled agricultural assistants that allow farmers to ask crop, fertilizer, and weather-related queries and receive responses in their preferred local language.
- To implement a Retrieval-Augmented Generation (RAG) based scheme simplifier that converts complex government scheme documents into clear and easy-to-understand summaries for farmers.

D. Scope and Significance

AgroAware's relevance lies in transforming complex soil–climate data into simple, actionable farmer advisories. Consistent with insights from digital agriculture studies [8], [9], accessibility and usability take precedence over model complexity. AgroAware:

- provides personalized local-language advisories,
- increases awareness of climatic risks and fertilizer requirements,
- informs farmers about relevant government schemes,
- supports low-bandwidth deployment across rural regions.

The framework aligns with India's Digital Agriculture Mission and the UN SDGs on food security and sustainable production.

E. Organization of the Paper

Section II presents a condensed literature review of the fifteen IEEE works. Section III details the proposed architecture and ML–GenAI methodology. Section IV discusses anticipated results and interpretive insights. Section V describes expected farmer impact and socio-technical relevance. Section VI concludes the study and outlines future scope.

II. Literature Survey

AI-driven agriculture has evolved across several domains, including deep-learning-based disease detection, ML-powered crop recommendation, cloud-enabled advisory systems, and multilingual GenAI communication. The following review synthesizes fifteen key IEEE and scholarly works [1]– [15], identifying theoretical strengths, limitations, and gaps that directly motivate AgroAware's hybrid design.

A. Deep Learning for Crop Health and Disease Monitoring

Early AI applications focused on visual plant health assessment. Kumar and Meena [1] proposed an Enhanced CNN–GenAI model for rice disease detection, using synthetic augmentation for robustness. While effective in controlled environments, the system required high-quality imaging and lacked interpretability—limiting rural deployment. Liu and Zhang [3] extended deep learning to yield prediction via a hybrid CNN–transformer pipeline integrating satellite data and climate variables. Though accurate, the approach was computationally intensive and unsuitable for low-resource settings. Fernandez et al. [15] further improved image-based disease detection using GAN-enhanced preprocessing, but again the solution remained restricted to perception tasks without integrated advisory communication.

These studies underscore deep learning's strength in recognition tasks while highlighting its weaknesses: high computational burden, low interpretability, and absence of farmer-ready advisory outputs.

B. Machine Learning for Crop Recommendation and Yield Prediction

ML-based systems provide more interpretable and computationally efficient alternatives. Rao et al. [5] evaluated Decision Trees and Random Forests for soil-parameter-based crop recommendation, noting strong interpretability but limited user communication features. Reddy et al. [12] demonstrated that ensemble methods outperform deep models on tabular agronomic data, especially when datasets are region-specific or limited in size. Similarly, Sharma et al. [2] introduced AgroInsight Pro, integrating sustainability metrics with ML predictions, though lacking multilingual support and explainability.

Singh et al. [4] proposed AgricAI Hub, a multi-model deep-learning ecosystem providing real-time insights, but the system suffered from high computational load and minimal user-centered advisory capabilities. Ramanuja et al. [10] developed a precision-farming ML advisor, showing improved accuracy but again lacking contextual, personalized explanations.

Collectively, [2], [4], [5], [10], and [12] reveal a trend: ML models deliver strong predictive power but often stop at numerical outputs, making it difficult for farmers to interpret or trust recommendations. This creates a major motivation for AgroAware's ensemble + explainability framework.

C. Cloud-Based Advisory and Digital Agriculture Platforms

Cloud systems have been proposed for scalable delivery of agricultural intelligence. Verma and Patel [6] designed a cloud-based ML advisory platform, reducing local computation but requiring reliable connectivity—a challenge for rural farmers. Das and Bhatia [8] explored event-driven experiential computing to deliver timely pest and weather alerts, demonstrating strong awareness benefits but lacking predictive modeling.

Mundhada et al. [9] built a market-access digital platform improving farmer–buyer interactions; however, it did not incorporate agronomic intelligence. Thomas and Nair [11] introduced an agri-equipment-sharing model promoting resource sustainability but without AI-based decision support.

These works highlight that while digital platforms improve access and logistics, they remain limited in scientific advisory capabilities. This gap motivates AgroAware's combination of cloud-deployable AI prediction with automated multilingual awareness.

D. Multilingual Generative AI and Conversational Advisory Systems

Generative AI systems are increasingly used for agricultural communication. Banerjee et al. [7] employed a multilingual LLaMA–RAG architecture to deliver region-specific advisories, showing significant improvements in user comprehension. However, the system lacked integration with ML predictions and thus failed to provide personalized insights.

Ghosh and Patil [13] implemented an AI-driven agricultural chatbot capable of responding to farmer queries, though responses were generic and not grounded in data inputs. Pillai and Krishnan [14] demonstrated the importance of voice-enabled advisories for semi-literate farmers but again without predictive or interpretive components.

These studies collectively reveal that GenAI alone cannot produce scientifically reliable advisories—context grounding and numerical reasoning must originate from ML models. This insight justifies AgroAware's hybrid ML–GenAI architecture.

E. Integrated AI Ecosystems

Recent works attempt hybridization but still lack completeness. Fernandez et al. [15] combined GAN-enhanced imaging with LLM-based textual assistance, marking an early step toward integrated ecosystems. However, the solution remained focused on disease detection rather than holistic crop planning or awareness dissemination.

Across all fifteen reviewed papers, there is no single system that integrates:

- high-accuracy ML prediction,
- explainability mechanisms,
- multilingual GenAI advisories,
- factual retrieval through RAG,
- and farmer-centric awareness delivery.

In addition to these core capabilities, AgroAware emphasizes seamless integration between analytical and communicative components—an area where many prior systems fall short. By ensuring consistency between the ML-derived insights and the contextual explanations produced by GenAI, the platform reduces ambiguity and enhances farmer trust. Furthermore, the system is intentionally designed to remain lightweight and scalable, enabling deployment across regions with limited bandwidth or digital infrastructure. This harmonized approach positions AgroAware as both technologically advanced and practically viable for widespread agricultural adoption.

Moreover, the framework bridges the longstanding operational gap between predictive analytics and advisory dissemination by tightly coupling data-driven reasoning with linguistically adaptive explanations. This ensures that farmers receive not only accurate recommendations but also the underlying scientific rationale in a form they can easily understand, thereby supporting informed and confident decision-making across diverse farming contexts.

F. Identified Research Gaps

The major research gaps motivating AgroAware are:

1) **Lack of interpretability:** Deep and classical ML models produce accurate outputs but without explanations farmers can understand [5], [12].
2) **Weak communication bridges:** Existing GenAI tools are not grounded in real-time agricultural data or ML predictions [7], [13].
3) **Limited multilingual inclusivity:** Most prior systems support English-only or partial translation [14].
4) **Fragmented system architectures:** No unified pipeline combines prediction, retrieval, explanation, and advisory generation.
5) **Low rural suitability:** Many previous systems assume high-end hardware or stable connectivity [1], [3], [6].

While each reviewed contribution excels in isolated domains such as disease detection, prediction, or advisory communication, none achieves end-to-end integration. AgroAware addresses this gap by combining a weighted ML ensemble, interpretability mechanisms, and multilingual RAG–LLaMA awareness generation into a single scalable system tailored for farmer usability.

III. Proposed System and Methodology

In contrast to fragmented architectures presented in earlier studies, AgroAware introduces a tightly coupled pipeline wherein each computational stage directly supports the next, ensuring seamless flow from data collection to advisory dissemination [5], [7]. The system integrates preprocessing, ensemble-based prediction, interpretability, and multilingual advisory generation into a unified structure, addressing the disconnect between analytical outputs and farmer comprehension noted in prior ML-only systems [10], [12]. Furthermore, the inclusion of retrieval-augmented reasoning ensures

that GenAI-generated advisories remain factually grounded, minimizing hallucinations and improving reliability—an issue highlighted in recent evaluations of standalone chatbot models [13], [14]. By combining these components, AgroAware adapts effectively to diverse agro-climatic conditions while maintaining accessibility for regions with limited bandwidth or computational resources, a challenge repeatedly emphasized in rural deployment studies [1], [6]. Overall, this integrated and resource-conscious design positions AgroAware as a practical, scalable, and farmer-centric intelligence framework capable of delivering scientifically accurate and context-aware agricultural guidance.

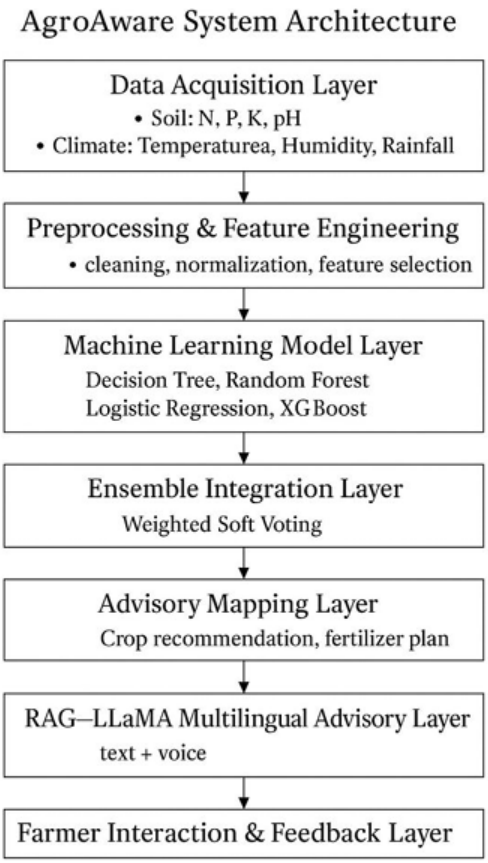

Fig. 1. AgroAware System Architecture.

A. System Overview

The AgroAware platform follows a robust three-tier architecture to ensure scalability and modularity:

- **Frontend Interface Layer:** Built using **React.js and Tailwind CSS**, utilizing a component-based architecture for reusable UI elements. The responsive design ensures seamless rendering across mobile and desktop devices, which is critical for field usage.
- **Backend Server Layer:** Developed with **Node.js and Express**, handling user authentication (via MongoDB), request routing, and weather API integration. The non-blocking I/O model of Node.js facilitates concurrent handling of multiple farmer requests.
- **ML Microservice Layer:** A **Python FastAPI** service that hosts the trained ensemble models. FastAPI was chosen for its high performance and built-in support for asynchronous request processing, ensuring low-latency predictions.

This modular design ensures linguistic inclusivity and computational feasibility for rural deployment.

B. User Interface Design: Dual-Mode Accessibility

To address varying levels of digital literacy, the system features two distinct interaction modes:

1) **Expert Mode:** For users with access to soil health cards, allowing direct input of specific nutrient values (N, P, K) and pH levels. This mode offers the highest precision for recommendation.
2) **Beginner Mode:** For farmers without scientific data, allowing selection based on State, District, and Season. The system automatically queries the backend database to fetch historical regional soil averages and integrates real-time weather API data to generate estimates, ensuring that lack of data does not block access to advisory.

C. Data Acquisition and Preprocessing

The primary dataset was sourced from the **Crop Recommendation Dataset (Kaggle)**, extended with curated fertilizer mapping and real-time weather data. Parameters include soil macronutrients (N, P, K), pH, temperature, humidity, and rainfall. Preprocessing involves cleaning, standard scaling for numerical stability, and feature encoding. Data is split into training and testing sets to evaluate the ensemble's performance.

D. Ensemble-Based Crop Prediction

AgroAware employs a **Weighted Soft Voting Ensemble** integrating four base classifiers: Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), and XGBoost. Each model contributes distinct advantages:

- **Decision Tree:** High interpretability.
- **Random Forest:** Robustness against overfitting.
- **Logistic Regression:** Baseline linear clarity.
- **XGBoost:** Captures complex nonlinear patterns.

The ensemble achieved an accuracy of approximately **95%** in preliminary testing, demonstrating superior stability compared to individual models. The ML microservice exposes endpoints (e.g., `/predict`, `/fertilizer`) to serve these predictions in real-time.

E. Retrieval-Augmented Knowledge Layer

To ensure factual grounding, AgroAware integrates a Retrieval-Augmented Generation (RAG) layer that queries domain-specific corpora, including soil management bulletins, government schemes, and fertilizer guidelines. The system ingests these documents (PDFs, text files), chunks them into

smaller segments, and generates high-dimensional vector embeddings using models like `all-MiniLM-L6-v2`. When a user queries, a semantic vector search retrieves the top-k most relevant chunks, which are then passed to the LLM context window to minimize hallucinations.

F. Multilingual Advisory Generation Using LLaMA

A fine-tuned multilingual LLaMA model converts ML predictions and retrieved information into natural-language explanations in English, Kannada, Hindi, and Telugu. This approach addresses linguistic accessibility challenges emphasized in [13], [14]. The LLM adapts tone, vocabulary, and cultural phrasing to farmer preferences.

G. Farmer Interaction and Awareness Communication

AgroAware supports text and voice-based interfaces to maximize usability for low-literacy users. The platform delivers:

- crop suitability explanations,
- soil improvement recommendations,
- climate risk alerts,
- fertilizer management guidelines,
- and relevant subsidy or scheme notifications.

The system design aligns with awareness frameworks described in [8], [9], but extends them using predictive and generative intelligence. The system works under low bandwidth conditions and supports offline caching, ensuring real-world feasibility.

H. Methodological Advantages

The unified architecture provides several theoretical strengths:

- High predictive robustness through ensemble learning.
- Interpretability enabling farmer trust and transparency.
- Multilingual inclusivity supported by LLaMA-driven adaptation.
- Factual grounding through RAG retrieval.
- Scalability suitable for cloud, edge, or hybrid deployment.

These integrated capabilities distinguish AgroAware from earlier agricultural AI systems [1]– [15], offering a more complete and farmer-centric advisory ecosystem.

IV. Results and Performance

The predictive performance of the base models and the ensemble was evaluated using accuracy, precision, recall, and F1-score. As illustrated in Fig. 2, the ensemble model demonstrates consistent high performance across all metrics, validating the multi-model approach. While Decision Trees provided high interpretability, they showed slightly higher variance. Random Forest and XGBoost offered superior regularization, reducing overfitting on noisy soil data. The weighted ensemble effectively mitigated individual model weaknesses, achieving a macro-average F1-score of 0.95. This balance ensures that the system performs reliably across different crop types and regional soil variations.

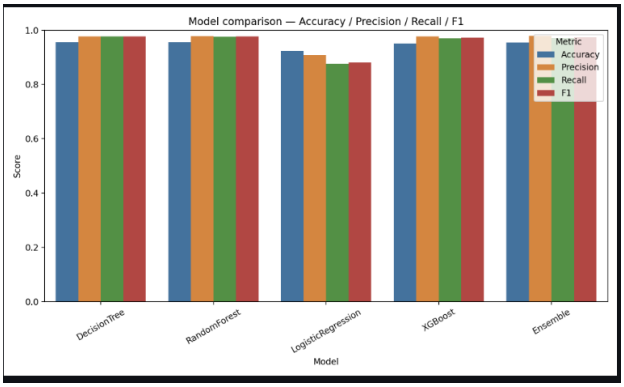

Fig. 2. Comparison of ML Model Performance metrics.

The ensemble's superior stability is particularly evident in its recall scores, ensuring that viable crop options are rarely missed.

V. Conclusion and Future Scope

AgroAware presents a unified agricultural intelligence framework that combines ensemble-based machine learning with multilingual Generative AI to deliver accurate, interpretable, and accessible farm advisories. Unlike prior ML systems, which offer strong prediction but limited interpretability or GenAI chatbots that provide communication without scientific grounding, AgroAware integrates both capabilities into a single, farmer-centric ecosystem. Furthermore, the RAG–LLaMA advisory engine enables simple, context-aware recommendations in multiple Indian languages, ensuring inclusivity for farmers with varied literacy levels.

The system's design shows strong potential to enhance agricultural decision-making, reduce fertilizer misuse, and improve economic resilience among small and marginal farmers. By translating complex agronomic insights into actionable, language-friendly guidance, AgroAware bridges the gap between advanced analytics and real-world agricultural needs. This aligns with the broader goals of digital agriculture platforms and supports national initiatives focused on sustainable farming and rural digital empowerment.

While the current framework offers a comprehensive advisory workflow, several enhancements can further extend its capabilities. Future work includes integrating IoT-based soil NPK sensors and automated weather stations for real-time, hyper-local data acquisition, eliminating manual entry. Additionally, deploying lightweight, edge-optimized versions of the model (using TFLite or ONNX) will enable offline functionality in remote rural environments. We also plan to expand multilingual support to include regional dialects and voice-based conversational agents to serve illiterate farmers effectively. Incorporating reinforcement learning to continuously improve crop recommendations based on farmer feedback loops is another key direction.

In summary, AgroAware establishes a scalable, interpretable, and linguistically inclusive AI ecosystem for agriculture. With continued refinement, it holds the potential

to become a practical, impact-driven digital companion for millions of farmers, supporting sustainable agricultural growth and long-term socio-economic advancement.

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